

```
1 function r = residual(za)
2
3     x0 = 0;
4     xf = 1;
5
6     y0 = 1;
7     yf_target = 0.5;
8     [~,y] = ode45(@dydx,[x0 xf],[y0; za]);
9     yf_calculated = y(end,1);
10    r = yf_calculated - yf_target;
11
12 end
```